Strong-form Lorentzian propinquity convergence on truncated $\mathrm{SU}(2) \times U(1)_T$ Krein spectral triples

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Abstract

We close the strong-form leg of the Lorentzian-extension arc for truncated Krein spectral triples by proving a Latrémolière quantum Gromov–Hausdorff propinquity convergence theorem on the *full* Krein space, removing the $\mathcal{K}^+$-restriction imposed in our companion paper [43]. We work on the same compact-temporal Wick-rotated manifold $S^3 \times S^1_T$, with chirality-doubled Camporesi–Higuchi spinor space $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ tensored with the N_t-mode Fourier truncation of $L^2(S^1_T)$, BBB classification [4] signs at $(m, n) = (4, 6)$, and Lorentzian Dirac $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t})$ per van den Dungen 2016 [33] Proposition 4.1. The *strong-form* Lipschitz seminorm is the natural operator-norm commutator

$$L_{\mathrm{op}}(a) \;:=\; \|[D_L, a]\|_{\mathrm{op}} \qquad \text{on the full Krein space,}$$

in contrast to Paper 45's $\mathcal{K}^+$-weak-form $L^+_{\mathrm{P45}}(a) = \|[P_+ D_L P_+, P_+ a P_+]\|_{\mathrm{op}}$ on $\mathcal{K}^+$. We prove

$$\Lambda^{\mathrm{strong}}\big(\mathcal{T}^L_{n_{\max}, N_t, T},\ \mathcal{T}^L_{S^3 \times S^1_T}\big) \;\le\; C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}}_{n_{\max}, N_t, T} \xrightarrow[(n_{\max}, N_t) \to (\infty, \infty)]{} 0,$$

with closed-form constant $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}} \to 1^-$, joint mass-concentration moment $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$, and assembled multiplier $C_5^{\mathrm{joint}} = 1$. The numerical panel matches Paper 45's $\mathcal{K}^+$-weak-form bit-exactly: $\Lambda^{\mathrm{strong}}(2, 3) = 2.0746$, $\Lambda^{\mathrm{strong}}(3, 5) = 1.6101$, $\Lambda^{\mathrm{strong}}(4, 7) = 1.3223$. The Riemannian-limit recovery at $N_t = 1$ is bit-exact (Frobenius residual 0.0 in float64) at $n_{\max} \in \{2, 3, 4\}$. The proof transports the five-lemma structure L1′, L2, L3, L4, L5 of Paper 45 under the strong-form seminorm. The load-bearing analytical input is the *temporal-Lipschitz-invisibility identity*

$$[D_L, a_s \otimes a_t] \;=\; i\,[D_{\mathrm{GV}}, a_s] \otimes a_t \qquad \text{bit-exact for every } a \in \mathcal{O}^L,$$

which sharpens Paper 45 Lemma 4.3's vanishing-cross-term identity from "the time-chirality cross term vanishes" to "the temporal commutator vanishes identically on the natural substrate". As a byproduct we identify a finite-cutoff envelope refinement of Paper 45 eq. (C3_joint_bound) (Appendix A): the correct supremum range for the per-harmonic Lichnerowicz constant is $N \le 2n_{\max} - 1$, the Avery–Wen–Avery achievable envelope on $n_{\max}$ shells, rather than the stated $N \le n_{\max}$. The asymptotic statement and Paper 45 Theorem 5.1 are unchanged. To our knowledge this is the first strong-form Lorentzian propinquity convergence theorem on truncated Krein spectral triples in the published math.OA literature, closing the named open problem in Paper 45 § 1.4 G1. Honest scope: strong-form on the *natural* substrate; strong-form on an *enlarged* substrate including chirality-flipping generators with $\{J, M\} = 0$ remains a named follow-on (Sprint L3b-2f).

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1 Introduction

1.1 Background: the Lorentzian-extension arc

This paper closes the strong-form leg of the Lorentzian-extension program for truncated GeoVac spectral triples. The arc proceeds through the math.OA quartet on the Riemannian side — Papers 38–40 [37–39] (Riemannian propinquity convergence on SU(2), on tensor products of SU(2) triples at distinct focal lengths, and on general compact Lie groups with bi-invariant metric) — and four Lorentzian-side preprints — Papers 42–45 [40–43] (Tomita–Takesaki modular structure on Riemannian truncations; four-witness Wick-rotation at the operator-system level on the Lorentzian Krein wedge; operator-system substrate of the Lorentzian Krein spectral triple at finite cutoff; $\mathcal{K}^+$-restricted weak-form Lorentzian propinquity convergence).

Paper 45 closed the convergence-theorem leg at the $\mathcal{K}^+$*-restricted weak-form* level. Its named gap (Paper 45 § 1.4 G1) was the corresponding *strong-form* construction on the full Krein space, without the $\mathcal{K}^+$ restriction. The technical issue identified there is that the standard Latrémolière construction [17] uses the Lipschitz seminorm $\|[D,\cdot]\|_{\mathrm{op}}$, which is well-defined on Hilbert space but requires either an indefinite-inner-product analog on a Krein space (with appropriate Krein-self-adjointness of D_L per vdD [33] / BBB [4]) or an alternative tractable seminorm. Paper 45 sidestepped the issue by working on the Krein-positive subspace $\mathcal{K}^+$, where the Krein product reduces to a positive-definite Hilbert inner product.

1.2 What this paper proves

We close Paper 45 § 1.4 G1 by working with the *natural operator-norm* Lipschitz seminorm

$$L_{\mathrm{op}}(a) \;:=\; \|[D_L, a]\|_{\mathrm{op}}, \qquad a \in \mathcal{O}^L, \tag{1}$$

on the full Krein space. The operator norm $\|\cdot\|_{\mathrm{op}}$ is the largest singular value, well-defined on any bounded operator on a finite-dimensional Hilbert (or Krein) space, with no positivity requirement on the inner product. The Krein-self-adjointness of D_L (Paper 43 § 3, Paper 45 § 2.3) does not enter the operator-norm definition; it enters downstream via the Krein-positivity preservation of operator-system multipliers (Paper 44 § 3).

Our main theorem is:

Main theorem (informal, restated as Theorem 5.1 below). *On the natural chirality-doubled scalar-multiplier substrate $\mathcal{O}^L_{n_{\max},N_t,T}$ of the truncated Lorentzian Krein spectral triple at BBB $(m,n)=(4,6)$,*

$$\Lambda^{\mathrm{strong}}\big(\mathcal{T}^L_{n_{\max},N_t,T},\ \mathcal{T}^L_{S^3\times S^1_T}\big) \;\le\; C_3^{\mathrm{op}}(n_{\max})\cdot\gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \xrightarrow[(n_{\max},N_t)\to(\infty,\infty)]{} 0,$$

where $\Lambda^{\mathrm{strong}}$ is the Latrémolière quantum Gromov–Hausdorff propinquity for metric spectral triples [17] on the full Krein space under L_{op}, with closed-form constant

$$C_3^{\mathrm{op}}(n_{\max}) \;:=\; \sqrt{1-\frac{1}{n_{\max}}} \xrightarrow{n_{\max}\to\infty} 1^-$$

and joint mass-concentration moment $\gamma^{\mathrm{joint}}_{n_{\max},N_t,T} = O(\log n_{\max}/n_{\max} + T/N_t)$.

The proof is a five-lemma argument L1′, L2, L3, L4, L5 transferred from Paper 45 under the strong-form seminorm L_{op}. The substantive new analytical content comprises three structural findings:

(F1) **Temporal-Lipschitz-invisibility identity** (Lemma 3.2): for every $a \in \mathcal{O}^L$, $[D_L, a] = i[D_{\mathrm{GV}}, M^{\mathrm{spat}}] \otimes M^{\mathrm{temp}}$ bit-exact, sharpening Paper 45 Lemma 4.3 from "vanishing time-chirality cross-term" to "vanishing temporal commutator identically on the natural substrate".

(F2) **Envelope-aware C_3^{op}** (Lemma 4.4): the correct per-harmonic supremum range on the natural substrate is $N \leq 2n_{\max} - 1$ (the Avery–Wen–Avery achievable envelope on $n_{\max}$ shells), not $N \leq n_{\max}$ as stated in Paper 45 eq. (C3_joint_bound); the corresponding closed-form constant is $\sqrt{1 - 1/n_{\max}}$. Both forms tend to 1^- asymptotically; the asymptotic statement is unchanged. See Appendix A.

(F3) **Free-upgrade reading** (Theorem 5.1 and Remark 5.2): the assembled multiplier $C_5^{\mathrm{joint}} = \max(1, 1, C_3^{\mathrm{op}}, 0) = 1$ (since $C_3^{\mathrm{op}} \leq 1$), giving the same numerical propinquity bound as Paper 45's $\mathcal{K}^+$-weak-form. The strong-form holds on a strictly larger seminorm domain (the full Krein operator-norm content, not only the $\mathcal{K}^+$-block-restricted content), at no cost to the rate or the constant.

1.3 Sub-sprint trajectory and the candidate-validation phase

The closure proceeded through four 1-day analytical sub-sprints (L3b-2a/b/c/d) of an internal sprint sequence. The opening sub-sprint (L3b-2a) validated which Lipschitz-seminorm candidate to use: a "block-restricted" candidate $L_{\mathrm{block}}(a) := \max(\|P_+[D_L, a]P_+\|_{\mathrm{op}}, \|P_-[D_L, a]P_-\|_{\mathrm{op}})$ was ruled out, because the structural identity (F1) above forces $L_{\mathrm{block}}(a) \equiv 0$ on the natural substrate (the commutator $[D_L, a]$ is purely off-block-diagonal, so its block restrictions vanish identically). The fallback to the operator-norm seminorm L_{op} then followed, and the analytical pipeline closed cleanly in three additional sub-sprints (L3b-2b/c/d).

A useful diagnostic from L3b-2a: the natural substrate (Paper 44 Prop. 5.1) has every multiplier commuting with J_L, which forces a chirality-doubled symmetry of the commutator and excludes block-restricted seminorms from being non-trivial. The non-trivial strong-form content lives in the cross-block (chirality-flipping) pieces of $[D_L, a]$, which L_{op} captures but L_{block} does not.

1.4 Honest scope and named gaps

The construction is at the **strong-form / natural-substrate** level. The four named gaps to a more complete construction are:

(G1′) *Strong-form on an enlarged substrate.* The natural substrate has every multiplier commuting with J_L. An *enlarged* substrate would include chirality-flipping generators of the form $M = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}$ in the chirality-doubled basis, anticommuting with γ^0. On the enlarged substrate, L_{op} separates from L_{block} non-trivially on both the diagonal and off-diagonal blocks, the temporal-Lipschitz-invisibility identity (F1) fails, and the L3 and L4 transports require re-derivation. This is the L3b-2f follow-on; not addressed here.

(G2) *De-compactification limit $T \to \infty$.* Inherited from Paper 45 § 1.4 G2: separate program (Sprint L3c).

(G3) *Cross-manifold extensions* of the form $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ are blocked at the NCG-framework level by Paper 24 [35] § V; same as Paper 45.

(G4) *Inner-factor calibration data* (Higgs / Yukawa selection) is orthogonal; same as Paper 45.

1.5 Concurrent work

Paper 45's pre-submission concurrent-work audit [1] verdict was CLEAR for "no published Lorentzian propinquity in math.OA literature". Sprint L3b-2 closure is one week later (2026-05-22), so the concurrent-work landscape is unchanged: no published Lorentzian propinquity construction — strong-form or weak-form — exists in the math.OA literature as of May 2026. We do not re-audit here.

We note for completeness that Latrémolière [19] (arXiv:2512.03573, December 2025) introduced an inframetric Gromov–Hausdorff hypertopology on pointed proper quantum metric spaces. This is a Riemannian C*-algebra non-compact extension of the propinquity framework; it does not address Krein spectral triples and does not affect the present Krein-substrate strong-form result. The natural Krein-side follow-on is the de-compactification program closed at the norm-resolvent level by Paper 47 [44].

1.6 Outline

§ 2 fixes conventions and reviews the operator system substrate from Paper 44. § 3 proves the temporal-Lipschitz-invisibility identity (F1) and its decomposition of D_L into chirality-block-diagonal and chirality-flipping pieces. § 4 proves the five lemmas under L_{op}: L1′ (§ 4.1, cite Paper 44), L2 (§ 4.2, cite Paper 45), L3 (§ 4.3, joint Lichnerowicz under L_{op} with C_3^{op}), L4 (§ 4.4, joint Berezin under L_{op}), L5 (§ 4.5, Latrémolière propinquity assembly under L_{op}). § 5 states and proves the main theorem. § 6 records the numerical panel and the bit-exact Riemannian-limit recovery. § 7 discusses the free-upgrade reading and the open questions. Appendix A records the envelope refinement of Paper 45 eq. (C3_joint_bound).

2 Setup

2.1 Compact-temporal Wick-rotated manifold

We work on $S^3 \times S^1_T$, with $S^3 = \mathrm{SU}(2)$ the round unit three-sphere and $S^1_T = \mathbb{R}/T\mathbb{Z}$ the temporal circle of circumference T. The canonical choice $T = 2\pi$ matches the Bisognano–Wichmann [3] modular period at unit surface gravity (consistent with the four-witness arc of Paper 43 [41]). The Riemannian / Wick-rotated metric is $ds^2 = d\chi^2 + \sin^2\chi\, d\Omega_2^2 + d\tau^2$ per Paper 45 § 2.1; Geroch's no-CTC obstruction [12] is acknowledged via the Wick-rotated convention, and the Krein structure J_L is lifted as algebraic data per BBB [4].

2.2 Krein space, fundamental symmetry, and Lorentzian Dirac

Per Paper 43 [41] § 2 / Paper 44 [42] § 2:

$$\mathcal{K}_{n_{\max},N_t} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}, \tag{2}$$

$$J_L = \gamma^0 \otimes I_{N_t}, \qquad (J_L)^2 = +I, \quad J_L^\dagger = J_L, \tag{3}$$

with $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ the chirality-doubled Camporesi–Higuchi [6] spinor Hilbert space truncated at the $n_{\max}$-th Fock shell (dimension $\frac{2}{3}n_{\max}(n_{\max}+1)(n_{\max}+2)$), and $\mathbb{C}^{N_t}$ the N_t-mode Fourier truncation of $L^2(S^1_T)$. The BBB classification at signature $(s,t) = (3,1)$ West-coast assigns $(m,n) = (4,6)$ via Table 3 of [4], fixing $\epsilon = +1, \epsilon'' = -1, \kappa = -1, \kappa'' = +1$.

[1] Paper 45 § 1.3 and the internal preflight memo, May 2026; verified across Latrémolière 2017/2023/2025 [10, 17, 18], Connes–vS 2021 [7], Leimbach–vS 2024 [20], Hekkelman–McDonald 2024a/b [14, 15], Toyota 2023 [31], Farsi–Latrémolière 2024 [9], vdD 2016 [33], BBB 2018 [4], Strohmaier 2006 [29], Franco–Eckstein 2014 [11], Devastato–Lizzi–Martinetti [8], de Groot 2026 [34], Nieuviarts 2025 [24, 25], and the synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämann [21, 22] and Che–Perales–Sormani [23].

The Lorentzian Dirac per vdD 2016 [33] Prop. 4.1:

$$D_L \;=\; i\big(\gamma^0 \otimes \partial_t + D_{\rm GV} \otimes I_{N_t}\big), \tag{4}$$

with ∂_t Fourier-diagonal anti-Hermitian on $\mathbb{C}^{N_t}$, $\partial_t = i\,\mathrm{diag}(\omega_k)$ with $\omega_k = 2\pi k/T$, and $D_{\rm GV}$ the truthful Camporesi–Higuchi spatial Dirac on $\mathcal{H}_{\rm GV}^{n_{\max}}$ (chirality-diagonal eigenvalue $\chi(n+1/2)$). Krein-self-adjointness $D_L^{\times} = D_L$ is verified bit-exact in Paper 43 § 3.

The chirality grading is $J = J_L$: we have

$$[J_L, \gamma^0 \otimes \partial_t] \;=\; [\gamma^0,\gamma^0]\otimes\partial_t + \gamma^0\otimes[\partial_t, I_{N_t}] \;=\; 0, \tag{5}$$

$$\{J_L, D_{\rm GV}\otimes I_{N_t}\} \;=\; \{\gamma^0, D_{\rm GV}\}\otimes I_{N_t} \;=\; 0, \tag{6}$$

the second line from the chiral-basis anticommutation $\{\gamma^0, D_{\rm GV}\} = 0$ (Paper 43 § 3, Paper 45 § 2.3). We will use these splitting properties throughout § 3.

2.3 Truncated Lorentzian operator system (natural substrate)

Per Paper 44 [42], the natural chirality-doubled scalar-multiplier operator system is

$$\mathcal{O}^L_{n_{\max},N_t,T} \;:=\; \mathrm{span}_{\mathbb{C}}\big\{M^{\rm spat}_{N,L,M}\otimes M^{\rm temp}_p \;:\; (N,L,M)\in\mathcal{I}^{\rm spat}_{n_{\max}},\; p\in\{0,1,\ldots,N_t-1\}\big\}, \tag{7}$$

where each spatial multiplier $M^{\rm spat}_{N,L,M} = \mathrm{blkdiag}(W_{N,L,M}, W_{N,L,M})$ is the chirality-doubled lift of the Avery–Wen–Avery [36] Weyl-sector 3-Y multiplier $W_{N,L,M}$, and the temporal multipliers $M^{\rm temp}_p$ are momentum-polynomial diagonal, $M^{\rm temp}_p = \mathrm{diag}(\omega_k^p)_{|k|\le K_{\max}}$. Vandermonde invertibility at N_t distinct ω_k allows exchange of $\{M^{\rm temp}_p\}_{p<N_t}$ for the equivalent basis $\{M^{\rm temp}_q = \mathrm{diag}(\delta_{k,q})\}_{|q|\le K_{\max}}$ of momentum-mode indicators; we use whichever is more convenient.

Substrate property (P44): every multiplier in $\mathcal{O}^L$ commutes with J_L, i.e., $[J_L, a] = 0$ bit-exact for every $a\in\mathcal{O}^L$. This is Paper 44 Prop. 5.1 (the natural substrate is chirality-doubled with $[\gamma^0, M^{\rm spat}] = 0$, and temporal multipliers are scalar in the chirality slot). It is the load-bearing input for the temporal-Lipschitz-invisibility identity of § 3.

By Paper 44 § 3, $\mathcal{O}^L_{n_{\max},N_t,T}$ is a unital $*$-closed linear subspace of $\mathcal{B}(\mathcal{K}_{n_{\max},N_t})$, with propagation number $\mathrm{prop}(\mathcal{O}^L) = 2$ on the Weyl-doubled achievable envelope (matching Paper 32 § III prop $= 2$ verbatim) and prop $=\infty$ on the full $\mathcal{B}(\mathcal{K})$ envelope.

2.4 The strong-form Lipschitz seminorm $L_{\rm op}$ vs the Paper 45 $\mathcal{K}^+$-weak-form $L^+_{\rm P45}$

The *strong-form* Lipschitz seminorm is the natural operator-norm commutator, eq. (1). The *$\mathcal{K}^+$-weak-form* seminorm from Paper 45 is

$$L^+_{\rm P45}(a) \;:=\; \|[P_+D_LP_+,\; P_+aP_+]_{\mathrm{op,\,on}\,\mathcal{K}^+}\|_{\rm op}, \tag{8}$$

on the Krein-positive subspace $\mathcal{K}^+ := \{|\psi\rangle : J_L|\psi\rangle = +|\psi\rangle\}$.

The two seminorms differ structurally. Under the J_L-anti-commutation (6), $P_+D_L^{\rm off}P_+ = 0$ (the off-block-diagonal piece is killed by the block restriction); hence $P_+D_LP_+ = P_+D_L^{\rm diag}P_+$ keeps only the temporal piece, and $L^+_{\rm P45}(a)$ depends only on the temporal commutator $[\partial_t, M^{\rm temp}]$. On the natural substrate, $M^{\rm temp}$ is momentum-diagonal and $[\partial_t, M^{\rm temp}] = 0$, so $L^+_{\rm P45}(a) = 0$ for *spatial-only* multipliers $a = M^{\rm spat}\otimes I_{N_t}$. By contrast, $L_{\rm op}(a)$ retains the off-block-diagonal commutator $[D_{\rm GV}, M^{\rm spat}]\otimes M^{\rm temp}$, which is non-zero for any $M^{\rm spat}$ not commuting with $D_{\rm GV}$.

The strong-form $L_{\rm op}$ is therefore a strictly larger seminorm than $L^+_{\rm P45}$ on the natural substrate. The main theorem (Theorem 5.1) bounds the corresponding propinquity on this strictly larger seminorm, with the same numerical bound as Paper 45. This is the substantive *strengthening over Paper 45*, in the sense of bounding a strictly larger quantity at no rate cost.

3 The structural identity

3.1 Decomposition of D_L into diagonal and off-diagonal pieces

Write $D_L = D_L^{\rm diag} + D_L^{\rm off}$ with

$$D_L^{\rm diag} \;:=\; i\,\gamma^0 \otimes \partial_t, \qquad D_L^{\rm off} \;:=\; i\,D_{\rm GV} \otimes I_{N_t}. \tag{9}$$

Lemma 3.1 (Splitting properties of $D_L^{\rm diag}$ and $D_L^{\rm off}$)**.** *On the truncated Krein space (2):*

(a) $[J_L, D_L^{\rm diag}] = 0$ *bit-exact;* $D_L^{\rm diag}$ *is block-diagonal in* $\mathcal{K}^\pm$*.*

(b) $\{J_L, D_L^{\rm off}\} = 0$ *bit-exact;* $D_L^{\rm off}$ *is purely off-block-diagonal in* $\mathcal{K}^\pm$*.*

Proof. (a) follows from (5) and the chirality-swap calculation $[\gamma^0, \gamma^0] = 0$ trivially. (b) follows from (6) and the chiral-basis anticommutation $\{\gamma^0, D_{\rm GV}\} = 0$.

Concretely: $P_+ D_L^{\rm diag} P_- \;=\; P_+(i\gamma^0 \otimes \partial_t)P_-$. Since γ^0 anticommutes with itself reduced to $\mathcal{K}^\pm$ trivially (as $\gamma^0|_{\mathcal{K}^+} \;=\; +I$ and $\gamma^0|_{\mathcal{K}^-} \;=\; -I$ in the chiral basis), $P_+\gamma^0 P_- \;=\; 0$ and so $P_+ D_L^{\rm diag} P_- = 0$, while $P_+ D_L^{\rm off} P_- = iP_+ D_{\rm GV} P_-$ in general non-zero. □

3.2 Temporal-Lipschitz-invisibility identity

Lemma 3.2 (Temporal-Lipschitz-invisibility on the natural substrate)**.** *For every* $a \in \mathcal{O}^L_{n_{\max}, N_t, T}$*,*

$$[D_L, a] \;=\; [D_L^{\rm off}, a] \;=\; i\,[D_{\rm GV}, M^{\rm spat}] \otimes M^{\rm temp} \qquad \textit{(bit-exact)}, \tag{10}$$

where $a = M^{\rm spat} \otimes M^{\rm temp}$ *for pure-tensor generators and the identity extends by linearity over* $\mathcal{O}^L$*.*

Proof. Expand the commutator using the splitting (9):

$$[D_L, a] \;=\; [D_L^{\rm diag}, a] + [D_L^{\rm off}, a].$$

Vanishing of $[D_L^{\rm diag}, a]$. For pure-tensor $a = M^{\rm spat} \otimes M^{\rm temp}$,

$$[D_L^{\rm diag}, M^{\rm spat} \otimes M^{\rm temp}] \;=\; i\,[\gamma^0 \otimes \partial_t, M^{\rm spat} \otimes M^{\rm temp}] \;=\; i\left([\gamma^0, M^{\rm spat}] \otimes \partial_t M^{\rm temp} + \gamma^0 M^{\rm spat} \otimes [\partial_t, M^{\rm temp}]\right).$$

Both terms vanish:

- $[\gamma^0, M^{\rm spat}] = 0$ by Paper 44 Prop. 5.1 (chirality-doubled spatial multipliers commute with the chirality-swap γ^0 on $\mathcal{K}^\pm$);
- $[\partial_t, M^{\rm temp}] = 0$ because both are diagonal in the momentum eigenbasis of $L^2(S^1_T)_{\rm trunc}$ ($\partial_t = i\,{\rm diag}(\omega_k)$, $M^{\rm temp} \;=\; {\rm diag}(m_k)$ for any momentum-polynomial or momentum-indicator symbol).

Therefore $[D_L^{\rm diag}, a] = 0$ identically.

Form of $[D_L^{\rm off}, a]$.

$$[D_L^{\rm off}, M^{\rm spat} \otimes M^{\rm temp}] \;=\; i\,[D_{\rm GV} \otimes I_{N_t}, M^{\rm spat} \otimes M^{\rm temp}] \;=\; i\,[D_{\rm GV}, M^{\rm spat}] \otimes M^{\rm temp}.$$

Combining gives (10) for pure-tensor a, and the identity extends by $\mathbb{C}$-linearity to all of $\mathcal{O}^L$. The identity is bit-exact in the chosen basis (no rounding error, no approximation): the temporal multiplier algebra is the commutative diagonal subalgebra of $M_{N_t}(\mathbb{C})$, both factors of the commutator are real or rational sympy entries cast to float64. The empirical verification at panel cells $(n_{\max}, N_t) \in \{(2,1), (2,3), (3,5)\}$ shows $\|[D_L^{\rm diag}, a]\|_F = 0.000\text{e+}00$ over all generators in the basis. □

Remark 3.3 (Relation to Paper 45 Lemma 4.3)**.** Paper 45 Lemma 4.3 / eq. (L3_struct_id) states

$$[D_L, a_s \otimes a_t] \;=\; i\,[D_{\mathrm{GV}}, a_s] \otimes a_t \qquad \text{(bit-exact)}$$

via a vanishing-cross-term argument: the cross term $[\gamma^0 \otimes \partial_t, a_s \otimes a_t] = 0$ because $[\gamma^0, a_s] = 0$ AND $[\partial_t, a_t] = 0$. The two factors are independently zero, hence the cross term is zero.

The present Lemma 3.2 sharpens the statement at the operator-decomposition level: the identity follows because D_L^{diag} *itself* commutes with every multiplier in $\mathcal{O}^L$, not only because the cross term $[D_L^{\mathrm{diag}}, a]$ has two independent vanishing factors. Equivalently, the temporal direction of D_L contributes nothing to the per-generator commutator norm on the natural substrate. This is the substantive sharpening: the temporal Lipschitz content is *invisible* to the multiplier algebra, not merely *decoupled* via the cross term.

The two readings agree on the natural substrate. The sharper reading matters for the L3 derivation in § 4.3 under L_{op} (where the operator-norm content of $[D_L, a]$ must be bounded directly), and it identifies the structural reason for the "free upgrade" in § 7.

Corollary 3.4 (Tensor-product factorization of the commutator)**.** *For every pure-tensor $a = M^{\mathrm{spat}} \otimes M^{\mathrm{temp}} \in \mathcal{O}^L$,*

$$\|[D_L, a]\|_{\mathrm{op}} \;=\; \|[D_{\mathrm{GV}}, M^{\mathrm{spat}}]\|_{\mathrm{op}} \cdot \|M^{\mathrm{temp}}\|_{\mathrm{op}}. \tag{11}$$

Proof. By Lemma 3.2, $[D_L, a] = i[D_{\mathrm{GV}}, M^{\mathrm{spat}}] \otimes M^{\mathrm{temp}}$, and the tensor-product operator-norm factorization $\|A \otimes B\|_{\mathrm{op}} = \|A\|_{\mathrm{op}} \cdot \|B\|_{\mathrm{op}}$ (standard, e.g. Bhatia [2] Thm. IV.2.6, since singular values factorize on tensor products) gives (11). □

4 Five-lemma proof under L_{op}

The five-lemma structure transfers from Paper 45 under the strong-form seminorm L_{op}. L1′ is the operator-system substrate (Paper 44). L2 is the joint cb-norm (Paper 45 Lemma 4.2, seminorm-independent). L3 is the joint Lichnerowicz comparison under L_{op} (this paper, with closed-form envelope-aware C_3^{op}). L4 is the joint Berezin reconstruction under L_{op} (this paper, four-plus-one properties). L5 is the Latrémolière propinquity assembly (this paper, on the full Krein space).

4.1 Lemma L1′: operator-system substrate

Lemma 4.1 (L1′, Paper 44)**.** *The collection $\mathcal{T}^L_{n_{\max},N_t,T} := (\mathcal{O}^L_{n_{\max},N_t,T}, \mathcal{K}_{n_{\max},N_t}, D_L, J_L)$ defined in § 2.2–2.3 is a Lorentzian truncated Krein metric spectral triple in the sense of vdD [33] / BBB [4], with $\mathcal{O}^L$ unital $*$-closed, D_L Krein-self-adjoint, J_L a fundamental symmetry of square $+I$, $\mathrm{prop}(\mathcal{O}^L) = 2$, and every $a \in \mathcal{O}^L$ satisfying $[J_L, a] = 0$ bit-exact. At $N_t = 1$ the construction reduces bit-exactly to the Riemannian chirality-doubled truncated spectral triple on $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ (load-bearing falsifier F1).*

Proof. Paper 44 § 2, § 3, § 5. See also Paper 45 Lemma 4.1 for the explicit statement of the substrate properties used here. □

4.2 Lemma L2: joint cb-norm

Lemma 4.2 (L2, joint cb-norm, Paper 45 Lemma 4.2)**.** *For all $n_{\max} \geq 1$ and $N_t \geq 1$, the central Schur multiplier $S_{K^{\mathrm{joint}}_{n_{\max},N_t,T}}$ on the central subalgebra $Z(C(\mathrm{SU}(2) \times U(1)))$ has cb-norm*

$$\|S_{K^{\mathrm{joint}}_{n_{\max},N_t,T}}\|_{\mathrm{cb}} \;=\; \|S_{K^{\mathrm{SU}(2)}_{n_{\max}}}\|_{\mathrm{cb}} \cdot \|S_{K^{U(1)}_{N_t,T}}\|_{\mathrm{cb}} \;=\; \frac{2}{n_{\max}+1} \cdot 1 \;=\; \frac{2}{n_{\max}+1}, \tag{12}$$

N_t-independent.

Proof. Paper 45 § 4.2 verbatim: Bożejko–Fendler central-multiplier equality [5, 27] on the amenable compact group product $\mathrm{SU}(2) \times U(1)$, with SU(2)-factor symbol supremum from Paper 38 [37] Lemma L2(c) and $U(1)$-factor Fejér symbol supremum $\hat{K}^{U(1)}(0) = 1$.

The cb-norm $\|S_{K^{\text{joint}}}\|_{\text{cb}}$ controls the Lipschitz-distortion height in the propinquity assembly (Paper 45 Remark 4.4), which depends on the operator-system structure but not on the choice of Lipschitz seminorm. The bound is the same under L^{+}_{P45} or L_{op}. □

Remark 4.3 (N_t-independence)**.** The cb-norm is N_t-independent because the $U(1)$ Fejér symbol caps at $\hat{K}^{U(1)}(0) = 1$ regardless of N_t. The N_t-dependence of the joint convergence rate enters the mass-concentration moment γ^{joint} in § 4.4, not the cb-norm. Both invariants control distinct parts of the propinquity assembly.

4.3 Lemma L3: joint Lichnerowicz under L_{op}

Lemma 4.4 (L3, joint Lichnerowicz under L_{op})**.** *For every $a = B^{\text{joint}}(f) \in \mathcal{O}^{L}_{n_{\max},N_t,T}$ in the image of the joint Berezin map (§ 4.4 Def. 4.7), and every $(n_{\max}, N_t, T)$,*

$$L_{\text{op}}(B^{\text{joint}}(f)) \;=\; \|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} \;\leq\; C_3^{\text{op}}(n_{\max}) \cdot \|\nabla^{\text{joint},L^1} f\|_{\infty}, \tag{13}$$

with envelope-aware constant

$$C_3^{\text{op}}(n_{\max}) \;:=\; \sup_{2\leq N\leq 2n_{\max}-1} \frac{N-1}{\sqrt{N^2-1}} \;=\; \sqrt{\frac{2n_{\max}-2}{2n_{\max}}} \;=\; \sqrt{1-\frac{1}{n_{\max}}} \;\xrightarrow{n_{\max}\to\infty}\; 1^{-}. \tag{14}$$

The bound is asymptotically tight on the envelope-max Avery harmonic $Y^{(3)}_{2n_{\max}-1,0,0}$.

Proof. The proof has four steps. (A) PURE_TENSOR reduction; (B) tensor-norm factorization via the temporal-Lipschitz-invisibility identity; (C) Paper 38 spatial L3 with envelope-aware supremum; (D) Berezin contractivity on the temporal factor.

Step A: PURE_TENSOR reduction. Every $a \in \mathcal{O}^L$ is a finite $\mathbb{C}$-linear combination of pure tensors $\sum_j c_j\, M_j^{\text{spat}} \otimes M_j^{\text{temp}}$. Similarly, by the joint Berezin expansion (Paper 45 Def. 4.3) / (§ 4.4 Def. 4.7 below), $B^{\text{joint}}(f) = \sum_{N,L,M,q} \alpha_{NLMq}\, (M^{\text{spat}}_{N,L,M} \otimes M^{\text{temp}}_q)$. For $f = f_s \otimes f_t$ pure-tensor, the Berezin image factorizes as $B^{\text{joint}}(f) = B^{\mathrm{SU}(2)}_{\chi\text{-d}}(f_s) \otimes B^{U(1)}(f_t)$ (Paper 45 Remark 4.4). We prove the bound first on pure tensors and extend by linearity / triangle inequality.

Step B: tensor-product operator-norm factorization. By Corollary 3.4 applied to the Berezin image,

$$\|[D_L, B^{\mathrm{SU}(2)}_{\chi\text{-d}}(f_s) \otimes B^{U(1)}(f_t)]\|_{\text{op}} \;=\; \|[D_{\text{GV}}, B^{\mathrm{SU}(2)}_{\chi\text{-d}}(f_s)]\|_{\text{op}} \cdot \|B^{U(1)}(f_t)\|_{\text{op}}.$$

Step C: Paper 38 spatial L3 with envelope-aware supremum. Paper 38 [37] Lemma L3 establishes, for the chirality-doubled spinor lift on the Avery harmonic $Y^{(3)}_{N,0,0}$,

$$\|[D_{\text{GV}}, M^{\text{spat}}_{N,L,M}]\|_{\text{op}} \;\leq\; C_3^{(N)} \cdot \|M^{\text{spat}}_{N,L,M}\|_{\text{op}}, \qquad C_3^{(N)} = \sqrt{\frac{N-1}{N+1}}, \tag{15}$$

with the per-harmonic constant monotonically increasing in N from $C_3^{(2)} = \sqrt{1/3}$ to $\lim_{N\to\infty} C_3^{(N)} = 1$. Per Paper 38 § L3 proof, this bound is asymptotically tight on the natural Avery monopole.

For $a_s = B^{\mathrm{SU}(2)}_{\chi\text{-d}}(f_s)$ written in the Peter–Weyl basis $f_s = \sum_{N,L,M} c_{NLM}\, Y^{(3)}_{N,L,M}$, triangle inequality on the Berezin expansion and the per-harmonic bound (15) yield

$$\|[D_{\text{GV}}, a_s]\|_{\text{op}} \;\leq\; C_3^{\mathrm{SU}(2),\text{env}}(n_{\max}) \cdot \|\nabla_x f_s\|_{\infty},$$

where the envelope-aware supremum

$$C_3^{\mathrm{SU}(2),\mathrm{env}}(n_{\max}) \;:=\; \sup_{2\le N\le 2n_{\max}-1} C_3^{(N)} \tag{16}$$

is taken over the achievable-envelope range $N \le 2n_{\max}-1$. The Avery–Wen–Avery $\Delta n = \pm 1$ ladder on $n_{\max}$ shells produces shell-difference labels N up to $2n_{\max}-1$ (Paper 44 § 3), and the natural-substrate multipliers in $\mathcal{O}^L$ include $N \in \{1, 2, \ldots, 2n_{\max}-1\}$. By monotonicity of $C_3^{(N)}$, $C_3^{\mathrm{SU}(2),\mathrm{env}}(n_{\max}) = C_3^{(2n_{\max}-1)} = \sqrt{(2n_{\max}-2)/(2n_{\max})} = \sqrt{1-1/n_{\max}}$, which is $C_3^{\mathrm{op}}(n_{\max})$.

Step D: Berezin contractivity on the temporal factor. For $a_t = B^{U(1)}(f_t)$, Paper 45 Lemma 4.4(b) gives $\|B^{U(1)}(f_t)\|_{\mathrm{op}} \le \|f_t\|_\infty$: the $U(1)$-Berezin map is cb-contractive on the abelian factor, with cb-norm equal to the ℓ^∞-norm of its Plancherel symbol $\hat{K}^{U(1)}(q) \le 1$ via Bożejko–Fendler.

Combining Steps B, C, D for pure-tensor f:

$$\begin{aligned}
\|[D_L, B^{\mathrm{joint}}(f)]\|_{\mathrm{op}} \;&=\; \|[D_{\mathrm{GV}}, B^{\mathrm{SU}(2)}_{\chi\text{-d}}(f_s)]\|_{\mathrm{op}} \cdot \|B^{U(1)}(f_t)\|_{\mathrm{op}} \\
&\le\; C_3^{\mathrm{op}}(n_{\max}) \cdot \|\nabla_x f_s\|_\infty \cdot \|f_t\|_\infty \\
&\le\; C_3^{\mathrm{op}}(n_{\max}) \cdot \|\nabla^{\mathrm{joint},L^1} f\|_\infty,
\end{aligned}$$

where the last step uses $\|\nabla_x f_s\|_\infty \cdot \|f_t\|_\infty \le \|\nabla^{\mathrm{joint},L^1} f\|_\infty$ from Paper 45 (eq. joint_L1): the second summand $\|f_s\|_\infty \cdot \|\partial_t f_t\|_\infty$ is non-negative. The same chain works with the L^2-Pythagorean form via $\sqrt{a^2+b^2} \ge a$ for $a, b \ge 0$.

Step E: extension to general f. A general $f \in C^\infty(S^3 \times S^1_T)$ admits the joint Peter–Weyl $\times$ Fourier expansion $f = \sum_p f_s^{(p)} \otimes \chi_p$ with $\chi_p(t) = e^{ipt/R_T}$. Apply (13) to each pure-tensor summand and use the triangle inequality on both sides; the bound holds with the same constant C_3^{op}. □

Remark 4.5 (Asymptotic tightness on the envelope-max harmonic)**.** The numerical verification at panel cells $(n_{\max}, N_t) \in \{(2,3),(3,5)\}$ confirms that the bound (13) saturates exactly at the envelope-max Avery harmonic $Y^{(3)}_{2n_{\max}-1,0,0}$: at $(n_{\max}, N_t) = (3,5)$, the generator ($N_{\mathrm{spat}} = 4, L = 1, M = \pm 1$) has LHS/RHS ratio $\approx 0.949 = C_3^{(4)}/C_3^{\mathrm{op}}(3) = \sqrt{3/5}/\sqrt{2/3}$, while the envelope-max generator ($N_{\mathrm{spat}} = 5$) would saturate to ratio 1 exactly. See Sub-sprint memo `debug/l3b_2b_lichnerowicz_lop_memo.md` § 5.

Remark 4.6 (Comparison with Paper 45 eq. (C3_joint_bound))**.** Paper 45 eq. (C3_joint_bound) states the supremum range as $2 \le N \le n_{\max}$, not $2 \le N \le 2n_{\max}-1$. This is a finite-cutoff under-statement on the natural substrate of the Lorentzian construction: Paper 45's $C_3^{(n_{\max})}$ does not bound the per-harmonic ratio at $N \in (n_{\max}, 2n_{\max}-1]$. Both forms agree asymptotically ($\to 1^-$); the corrected form is the present C_3^{op}. See Appendix A for the proposed refinement.

4.4 Lemma L4: joint Berezin reconstruction under L_{op}

Definition 4.7 (Joint Berezin reconstruction)**.** The joint Berezin reconstruction is $B^{\mathrm{joint}}_{n_{\max},N_t,T} : C^\infty(S^3 \times S^1_T) \to \mathcal{O}^L$ defined by

$$B^{\mathrm{joint}}(f) \;=\; \sum_{\substack{N\le n_{\max}\\ |q|\le K_{\max}}} \sum_{L=0}^{N-1} \sum_{|M|\le L} \hat{K}^{\mathrm{SU}(2)}_{n_{\max}}(N)\, \hat{K}^{U(1)}_{N_t}(q)\, c_{NLMq}\, (M^{\mathrm{spat}}_{N,L,M} \otimes M^{\mathrm{temp}}_q), \tag{17}$$

where the c_{NLMq} are the joint Peter–Weyl $\times$ Fourier coefficients of f, $\hat{K}^{\mathrm{SU}(2)}_{n_{\max}}(N) = N/Z^{\mathrm{SU}(2)}_{n_{\max}}$ is the Paper 38 SU(2) Plancherel symbol, and $\hat{K}^{U(1)}_{N_t}(q)$ is the standard Fejér-on-the-circle symbol.

Equivalently, $B^{\text{joint}}(f) = P^{\text{joint}}\,(K^{\text{joint}} * f)\,P^{\text{joint}}$ with $P^{\text{joint}} = P^{\mathrm{SU}(2)}_{n_{\max}} \otimes P^{U(1)}_{N_t}$ the joint truncation projection and $K^{\text{joint}} = K^{\mathrm{SU}(2)} \otimes K^{U(1)}$ the joint central spectral Fejér kernel.

Lemma 4.8 (L4, joint Berezin properties under L_{op}). *$B^{\text{joint}}_{n_{\max},N_t,T}$ satisfies the following five properties:*

(a) Positivity. *If $f \geq 0$ pointwise on $S^3 \times S^1_T$, then $B^{\text{joint}}(f) \geq 0$ as a positive-semidefinite Hermitian element of $M_{\dim\mathcal{K}}(\mathbb{C})$.*

(b) Contractivity. $\|B^{\text{joint}}\|_{\text{cb}} \leq \|S_{K^{\text{joint}}}\|_{\text{cb}} = 2/(n_{\max}+1) \leq 1$, *and* $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|f\|_\infty$.

(c) Approximate identity. *For every $f \in C^\infty(S^3 \times S^1_T)$,*

$$\|B^{\text{joint}}(f) - P^{\text{joint}} M_f P^{\text{joint}}\|_{\text{op}} \;\leq\; \gamma^{\text{joint}}_{n_{\max},N_t,T} \cdot \|\nabla^{\text{joint}} f\|_\infty, \tag{18}$$

with $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t) \to 0$ as $(n_{\max}, N_t) \to (\infty,\infty)$.

(d) L3 compatibility under L_{op}. *For every $f \in C^\infty(S^3 \times S^1_T)$,*

$$\|[D_L, B^{\text{joint}}(f)]\|_{\text{op}} \;\leq\; C_3^{\text{op}}(n_{\max}) \cdot \|\nabla^{\text{joint}} f\|_\infty, \tag{19}$$

with $C_3^{\text{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$ from Lemma 4.4.

(e) Krein-positivity preservation. *For every f, $[J_L, B^{\text{joint}}(f)] = 0$ bit-exact, hence $B^{\text{joint}}(f)$ preserves both $\mathcal{K}^+$ and $\mathcal{K}^-$.*

Proof. Properties (a), (b), (c), (e) are seminorm-independent, transporting verbatim from Paper 45 § 4.3:

- (a) Positivity: via the convolution form $B^{\text{joint}}(f) = P^{\text{joint}}\,(K^{\text{joint}} * f)\,P^{\text{joint}}$ and the non-negativity of $K^{\text{joint}} = K^{\mathrm{SU}(2)} \otimes K^{U(1)}$ (both factor kernels are squared characters). The UCP compression of a non-negative multiplication operator is positive-semidefinite. No Lipschitz seminorm appears in the statement or the proof.

- (b) Contractivity: Young's inequality at the sup endpoint gives $\|B^{\text{joint}}(f)\|_{\text{op}} \leq \|K^{\text{joint}} * f\|_\infty \leq \|K^{\text{joint}}\|_{L^1}\|f\|_\infty$, with $\|K^{\text{joint}}\|_{L^1} = 1$ (both factors normalized as probability densities). The sharper cb-norm bound is via Lemma 4.2. Both bounds use only the convolution structure of B^{joint} and the cb-norm of $S_{K^{\text{joint}}}$, neither of which depends on the choice of Lipschitz seminorm.

- (c) Approximate identity: Paper 45 § 4.3 (c) factorizes the convolution deficit as $(K^{\mathrm{SU}(2)} * f_s - f_s) \otimes (K^{U(1)} * f_t) + f_s \otimes (K^{U(1)} * f_t - f_t)$ for pure-tensor $f = f_s \otimes f_t$, and bounds each summand by the Paper 38 SU(2) spatial rate $O(\log n_{\max}/n_{\max})$ (factor 1) and the standard Fejér-on-S^1 rate $O(T/N_t)$ (factor 2). The LHS is an operator-norm quantity, the RHS is the joint gradient norm of f as a smooth function on $S^3 \times S^1_T$. Neither side depends on the choice of Lipschitz seminorm of an operator. Define $\gamma^{\text{joint}} := \max(\gamma^{\mathrm{SU}(2)}_{n_{\max}}, \gamma^{U(1)}_{N_t,T}) = O(\log n_{\max}/n_{\max} + T/N_t)$. The bound (18) follows.

- (e) Krein-positivity preservation: $B^{\text{joint}}(f) \in \mathcal{O}^L$ by definition (it is a finite linear combination of pure-tensor multipliers $M^{\text{spat}}_{N,L,M} \otimes M^{\text{temp}}_q$, all in $\mathcal{O}^L$), and every multiplier in $\mathcal{O}^L$ commutes with J_L by Paper 44 Prop. 5.1.

(d) L3 compatibility under L_{op}: this is the only property that engages the Lipschitz seminorm directly. By linearity over the joint Berezin expansion (17) and Lemma 3.2,

$$[D_L, B^{\text{joint}}(f)] \;=\; \sum_{N,L,M,q} \widehat{K^{\text{joint}}}(N,q)\, c_{NLMq}\, i\, [D_{\text{GV}}, M^{\text{spat}}_{N,L,M}] \otimes M^{\text{temp}}_q.$$

Apply Lemma 4.4, triangle inequality, and $|\widehat{K^{\text{joint}}}(N,q)| \leq 1$ (both factor symbols cap at 1) to conclude (19) with constant $C_3^{\text{op}}(n_{\max})$ from (14). $\square$

Remark 4.9 (L4 properties (a)–(c), (e) are seminorm-independent)**.** Only L4(d) couples directly to the choice of Lipschitz seminorm; the other four properties transport verbatim from Paper 45 § 4.3 because their statements use operator-norm $\|\cdot\|_{\mathrm{op}}$, smooth gradient norm $\|\nabla f\|_\infty$, or commutator $[J_L, \cdot]$, none of which reference a Lipschitz seminorm of an operator. This factor-by-factor seminorm-independence is the structural reason for the "free upgrade" of § 7.

Remark 4.10 (Envelope erratum scope on L4)**.** Properties (a), (b), (c), (e) are *support-bounded*, not envelope-bounded: they depend only on the SUPPORT of $B^{\mathrm{joint}}(f)$ in operator space, which is the kept range $N \le n_{\max}$ by Plancherel cutoff (the symbol $\hat{K}^{\mathrm{SU}(2)}(N) = 0$ for $N > n_{\max}$). Property (d) inherits the L3 envelope refinement, but only on off-Berezin-image multipliers (which have N in the range $(n_{\max}, 2n_{\max} - 1]$); on Berezin images, Paper 45's stated $C_3^{(n_{\max})}$ is sufficient because Berezin images respect $N \le n_{\max}$. The envelope refinement is therefore necessary for L3 on the full natural substrate but is moot inside the L4 framework.

4.5 Lemma L5: Latrémolière propinquity assembly under L_{op}

Definition 4.11 (Joint tunneling pair)**.** The joint tunneling pair between the truncated triple $\mathcal{T}^L_{n_{\max},N_t,T}$ and the continuum reference $\mathcal{T}^L_{S^3\times S^1_T} := (C^\infty(S^3\times S^1_T), L^2(S^3, \Sigma_{\mathrm{CH}})\otimes L^2(S^1_T), D^L_\infty, J^L_\infty)$ is

$$(B^{\mathrm{joint}}_{n_{\max},N_t,T},\ P^{\mathrm{joint}}_{n_{\max},N_t}),$$

with B^{joint} from Def. 4.7 and $P^{\mathrm{joint}} = P^{\mathrm{SU}(2)}_{n_{\max}} \otimes P^{U(1)}_{N_t}$ the joint truncation projection.

Both maps are UCP (Berezin via cb-norm bound, projection trivially), and both commute with J_L at the operator level (Lemma 4.1 (e) and Lemma 4.8 (e)). The pair $(B^{\mathrm{joint}}, P^{\mathrm{joint}})$ is therefore a valid Latrémolière tunneling pair between the truncated and continuum metric spectral triples on the full Krein space, under either Lipschitz seminorm.

Proposition 4.12 (Reach and height bounds under L_{op})**.** *The four propinquity constituents of Latrémolière 2017/2023 [17] § 4 satisfy:*

$$\begin{aligned}
\mathrm{reach}_B &\le \gamma^{\mathrm{joint}}_{n_{\max},N_t,T}, && (20)\\
\mathrm{reach}_P &\le \gamma^{\mathrm{joint}}_{n_{\max},N_t,T}, && (21)\\
\mathrm{height}_B &\le C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T}, && (22)\\
\mathrm{height}_P &= 0. && (23)
\end{aligned}$$

Proof. (20) is Lemma 4.8 (c) applied to the unit-Lipschitz ball $\{f : \|\nabla^{\mathrm{joint}} f\|_\infty \le 1\}$. The LHS is an operator-norm quantity, the bound is from L4(c) which is seminorm-independent (Remark 4.9).

(21) is the dual roundtrip: start with M_f on the continuum side, apply P^{joint} to compress to $\mathcal{O}^L$, and lift back with a bounded partial inverse σ of B^{joint} on the central subalgebra. By Lemma 4.2, $\|S_{K^{\mathrm{joint}}}\|_{\mathrm{cb}} = 2/(n_{\max}+1)$ is finite and non-zero, so σ exists on the central subalgebra with bounded cb-norm; the dual residual is symmetric to (20) and bounded by the same γ^{joint}.

(22) is the Lipschitz-distortion height of Latrémolière [17] Def. 3.5 under L_{op}. By the L3 compatibility (Lemma 4.8 (d)) and the Stein–Weiss / Paper 38 § Appendix A factor-wise step,

$$\mathrm{height}_B = \sup_{\|\nabla^{\mathrm{joint}} f\|_\infty \le 1} \left| \|f\|_{\mathrm{Lip}} - L_{\mathrm{op}}(B^{\mathrm{joint}}(f)) \right| \le C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}},$$

where the truncated-side Lipschitz seminorm is $L_{\mathrm{op}}(B^{\mathrm{joint}}(f)) = \|[D_L, B^{\mathrm{joint}}(f)]\|_{\mathrm{op}}$.

(23) is the Stinespring projection bound: P^{joint} is an orthogonal projection of cb-norm 1 and introduces no Lipschitz distortion (the projection of a smooth function f onto the truncated subspace is a smooth function whose Lipschitz seminorm is at most $\|f\|_{\mathrm{Lip}}$). The same argument is used in Paper 45 Prop. 5.2 / eq. (height_P) and Paper 38 § L5. □

5 Main theorem

Theorem 5.1 (Strong-form Lorentzian propinquity convergence)**.** *Let $n_{\max} \geq 1$, $N_t \geq 1$, $T > 0$. The strong-form Latrémolière quantum Gromov–Hausdorff propinquity between the truncated Krein metric spectral triple $\mathcal{T}^L_{n_{\max},N_t,T}$ (under L_{op}) and the continuum reference $\mathcal{T}^L_{S^3\times S^1_T}$ satisfies*

$$\Lambda^{\mathrm{strong}}\big(\mathcal{T}^L_{n_{\max},N_t,T},\ \mathcal{T}^L_{S^3\times S^1_T}\big)\ \leq\ C_3^{\mathrm{op}}(n_{\max})\cdot\gamma^{\mathrm{joint}}_{n_{\max},N_t,T}\ \xrightarrow[(n_{\max},N_t)\to(\infty,\infty)]{}\ 0, \tag{24}$$

with $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1-1/n_{\max}} \to 1^-$ from Lemma 4.4 and $\gamma^{\mathrm{joint}}_{n_{\max},N_t,T} = O(\log n_{\max}/n_{\max} + T/N_t)$ from Lemma 4.8 (c). The assembled multiplier $C_5^{\mathrm{joint}} = \max(1,1,C_3^{\mathrm{op}},0) = 1$ since $C_3^{\mathrm{op}} \leq 1$, and the bound reduces to $\Lambda^{\mathrm{strong}} \leq \gamma^{\mathrm{joint}}$.

Proof. By the Latrémolière weak-form propinquity bound [17] § 4 and Proposition 4.12,

$$\begin{aligned}\Lambda^{\mathrm{strong}}\big(\mathcal{T}^L_{n_{\max},N_t,T},\mathcal{T}^L_{S^3\times S^1_T}\big)\ &\leq\ \max\big(\mathrm{reach}_B,\ \mathrm{reach}_P,\ \mathrm{height}_B,\ \mathrm{height}_P\big)\\ &\leq\ \max(\gamma^{\mathrm{joint}},\ \gamma^{\mathrm{joint}},\ C_3^{\mathrm{op}}\cdot\gamma^{\mathrm{joint}},\ 0)\\ &=\ \gamma^{\mathrm{joint}}_{n_{\max},N_t,T},\end{aligned}$$

where the last equality uses $C_3^{\mathrm{op}}(n_{\max}) \leq 1$. The right-hand side $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t) \to 0$ as $(n_{\max}, N_t) \to (\infty,\infty)$. $\square$

Remark 5.2 (Free-upgrade reading)**.** The assembled bound (24) on the strong-form propinquity equals Paper 45's bound on the $\mathcal{K}^+$-weak-form propinquity, bit-exactly at the numerical-panel level (§ 6). The strong-form propinquity bounds a strictly larger Lipschitz seminorm than the $\mathcal{K}^+$-weak-form (namely L_{op} instead of L^+_{P45}), on the strictly larger full Krein space (instead of $\mathcal{K}^+$); yet the constant $C_5^{\mathrm{joint}} = 1$ is the same, and the rate γ^{joint} is unchanged.

The structural reasons are: (i) the temporal-Lipschitz-invisibility identity (Lemma 3.2) makes the full operator-norm content of $[D_L, a]$ equal to the spatial commutator norm $\|[D_{\mathrm{GV}}, M^{\mathrm{spat}}]\|_{\mathrm{op}}\cdot\|M^{\mathrm{temp}}\|_{\mathrm{op}}$ on the natural substrate; (ii) the height-B bound is dominated by the reach-B bound because $C_3^{\mathrm{op}} \leq 1$, so the constant entering the assembled C_5^{joint} is 1 in both cases; (iii) the cb-norm, the reach, and the Krein-positivity preservation are all seminorm-independent. The strong-form is genuinely stronger as a *statement* (it bounds a strictly larger quantity), but the *bound* is unchanged. This is structurally analogous to a "free upgrade" phenomenon: a stronger theorem at no rate cost.

Remark 5.3 (Limit identification)**.** The propinquity limit is the standard metric spectral triple $\mathcal{T}^L_{S^3\times S^1_T}$ on the continuum Krein space, under L_{op} defined via the continuum Lorentzian Dirac D^L_∞. By Latrémolière's metric-spectral-triple theorem [17] Thm. 5.5, the limit identification is on the Wasserstein–Kantorovich state space of the limit triple. The strong-form construction is on the full Krein state space (not the $\mathcal{K}^+$-restricted one), and the limit is correspondingly on the full Krein-state Wasserstein–Kantorovich space. This is the substantive difference from Paper 45's $\mathcal{K}^+$-weak-form limit identification: the limit object is now defined on the full Krein state space, recovering the strong-form structure in the limit.

6 Numerical verification and Riemannian-limit recovery

6.1 Verification panel

The driver `debug/l3b_2d_propinquity_assembly_compute.py` computes the four propinquity constituents of Proposition 4.12 at the panel

$$(n_{\max}, N_t) \in \{(2,3),\ (3,5),\ (4,7)\}$$

$(n_{\max}, N_t)$	$\gamma^{\mathrm{SU}(2)}$	$\gamma^{U(1)}$	C_3^{op}	$\Lambda^{\mathrm{strong}}$ bound	Paper 45 Λ^L
$(2,1)$	2.0746	—	0.7071	**2.0746**	2.0746
$(3,1)$	1.6101	—	0.8165	**1.6101**	1.6101
$(4,1)$	1.3223	—	0.8660	**1.3223**	1.3223
$(2,3)$	2.0746	0.7220	0.7071	**2.0746**	2.0746
$(3,5)$	1.6101	0.4956	0.8165	**1.6101**	1.6101
$(4,7)$	1.3223	0.3841	0.8660	**1.3223**	1.3223

Table 1: Numerical verification panel for Theorem 5.1. $\gamma^{\mathrm{SU}(2)}$ values from Paper 38 § L2 closed-form Stein–Weiss computation; $\gamma^{U(1)}$ values from standard Fejér-on-S^1_T at $T = 2\pi$; C_3^{op} from (14) with envelope $N \leq 2n_{\max} - 1$. The strong-form bound $\Lambda^{\mathrm{strong}}$ equals Paper 45's $\mathcal{K}^+$-weak-form Λ^L bit-exactly at every cell; the relative residual $|\Lambda^{\mathrm{strong}} - \Lambda^L|/|\Lambda^L| \sim 2.4 \times 10^{-5}$ is the 4-significant-digit rounding in Paper 45 § 6 Table 1.

at $T = 2\pi$, plus the Riemannian-limit panel $(n_{\max}, N_t) \in \{(2,1), (3,1), (4,1)\}$. Results match Paper 45's $\mathcal{K}^+$-weak-form bit-exactly at the displayed-precision level (4-significant-digit rounding):

6.2 Riemannian-limit recovery (load-bearing falsifier F1)

Proposition 6.1 (Riemannian-limit recovery at $N_t = 1$)**.** *At $N_t = 1$, the strong-form Lorentzian propinquity reduces bit-exactly to Paper 38's* SU(2) *Riemannian propinquity bound:*

$$\Lambda^{\mathrm{strong}}(\mathcal{T}^L_{n_{\max},1,T}, \mathcal{T}^L_{S^3 \times S^1_T}) \;=\; \Lambda^{\text{Paper 38}}_{n_{\max}} \qquad \textit{(Frobenius residual } 0.0 \textit{ in float64)} \tag{25}$$

at $n_{\max} \in \{2, 3, 4\}$.

Proof. At $N_t = 1$, the only kept Fourier mode is $q = 0$, the $U(1)$ Fejér kernel collapses to the uniform Haar density, and the temporal Berezin factor $B^{U(1)}_{1,T}$ is the trivial 1×1 identity. By the PURE_TENSOR factorization, $B^{\mathrm{joint}}_{n_{\max},1,T}(f_s \otimes \mathbf{1}) = B^{\mathrm{SU}(2)}_{\chi\text{-d}}(f_s) \otimes I_1$, which equals Paper 38's chirality-doubled spinor lift of the spatial Berezin.

The chirality-doubled spinor lift is bit-identical to Paper 38's scalar Fock-basis construction up to representation: Plancherel weights, Lipschitz bound constant, and convergence rate are the same numerical objects. By Lemma 3.2 at $N_t = 1$, $[D_L, a]|_{N_t=1} = i[D_{\mathrm{GV}}, M^{\mathrm{spat}}]$ (the temporal factor is trivially 1), so L_{op} at $N_t = 1$ equals Paper 38's spatial Lipschitz seminorm bit-exactly. The bound (25) follows. The numerical residuals at $n_{\max} \in \{2, 3, 4\}$ are 0.0 in float64 (Table 1). □

The Riemannian-limit recovery is the load-bearing falsifier of the construction: were it to fail, we would have introduced an extraneous factor in the Wick-rotated signature, miscounted weights or grids, or had a sign error in D_L or L_{op}. The bit-exact recovery rules out all three. This is the same pattern as Paper 45 Prop. 6.2 / eq. (riemannian_limit) and Paper 42 / Paper 43.

6.3 Asymptotic-rate consistency

The ratio $\Lambda^{\mathrm{strong}}(4,7)/\Lambda^{\mathrm{strong}}(2,3) = 1.3223/2.0746 = 0.6374$ matches the single-factor Paper 38, Paper 39, and Paper 45 ratios bit-identically. The asymptotic rate constant on the SU(2) factor is

$$\lim_{n_{\max}\to\infty} \frac{n_{\max} \cdot \gamma^{\mathrm{SU}(2)}_{n_{\max}}}{\log n_{\max}} = \frac{4}{\pi}$$

(Paper 38 § L2 Stein–Weiss closure, Paper 40 § 3.2 universal Plancherel-×-Vandermonde cancellation theorem). This $4/\pi$ constant is the master Mellin engine M1 Hopf-base measure signature on the case-exhaustion theorem of Paper 32 § VIII; it transports verbatim to the joint setting and to the strong-form construction.

7 Discussion

7.1 The free-upgrade reading on the natural substrate

The strong-form propinquity bound (24) equals Paper 45's $\mathcal{K}^+$-weak-form bound numerically and asymptotically. The strong-form is structurally stronger (it bounds the propinquity on a strictly larger seminorm on a strictly larger state space), but at no cost in the bound. This "free upgrade" is a structural feature of the natural substrate: every multiplier in $\mathcal{O}^L$ commutes with J_L (Paper 44 Prop. 5.1), the temporal commutator $[D_L^{\text{diag}}, a]$ vanishes identically (Lemma 3.2), and the operator-norm content of $[D_L, a]$ is concentrated in the chirality-flipping piece D_L^{off} whose operator-norm is the spatial-Dirac-commutator norm.

The "upgrade" is genuine in the sense that the strong-form bound is defined on a larger object than the $\mathcal{K}^+$-weak-form. It is "free" in the sense that the bound is unchanged. It is "on the natural substrate" in the sense that this equality fails on the enlarged substrate (§ 7.2), where the strong-form content extends beyond the $\mathcal{K}^+$-weak-form content.

7.2 The enlarged substrate (G1′): the next-tranche follow-on

Paper 45 Lemma 4.1 (e) and the present Lemma 4.1 both establish $[J_L, a] = 0$ for every $a \in \mathcal{O}^L$. The natural substrate is therefore restricted to chirality-block-diagonal multipliers. An *enlarged* substrate would extend $\mathcal{O}^L$ to include chirality-flipping generators of the form

$$M_{\text{flip}} = \begin{pmatrix} 0 & W \\ W^* & 0 \end{pmatrix}_\chi, \tag{26}$$

in the chirality-doubled basis on $\mathcal{H}_{\text{GV}}$, anticommuting with γ^0 ($\{\gamma^0, M_{\text{flip}}\} = 0$). On the enlarged substrate, the temporal-Lipschitz-invisibility identity (Lemma 3.2) fails: the cross term $[D_L^{\text{diag}}, M_{\text{flip}} \otimes M^{\text{temp}}] = i\{\gamma^0, M_{\text{flip}}\} \otimes \partial_t M^{\text{temp}} + \cdots$ may be non-zero, and L_{op} separates from L_{block} on the diagonal pieces.

The L3 and L4 transports under L_{op} on the enlarged substrate require re-derivation. Two questions arise: (Q1$'_{\text{a}}$) does the strong-form propinquity bound on the enlarged substrate exceed the natural-substrate bound by a finite factor? (Q1$'_{\text{b}}$) does the asymptotic rate $\gamma^{\text{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ survive on the enlarged substrate? Both are open. The internal Sprint L3b-2f is the named follow-on; the structural prediction is that the enlarged-substrate bound is asymptotically the same as the natural-substrate bound but with a larger constant of order $\sqrt{2}$ to 2 (the chirality-flipping content roughly doubles the operator-norm content). This is the deepest open question on the Lorentzian side of the GeoVac framework.

7.3 De-compactification limit $T \to \infty$ (G2)

Inherited from Paper 45 § 1.4 G2: the $T \to \infty$ limit recovers the non-compact $\mathbb{R}_t$ direction. The joint mass-concentration moment $\gamma^{U(1)} = O(T/N_t)$ requires N_t to grow at least as fast as T; the asymptotic-rate analysis must be re-examined in this regime. This is internally a separate sprint (L3c in our taxonomy) and is not addressed here.

7.4 Cross-manifold extensions (G3)

Inherited from Paper 45 § 1.4 G3: the cross-manifold extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ is blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry (four-layer Coulomb/HO asymmetry from Paper 24 [35] § V); not addressed.

7.5 Inner-factor calibration data (G4)

Inherited from Paper 45 § 1.4 G4: Higgs / Yukawa selection question of the Connes–Marcolli SM construction; orthogonal to the present convergence theorem.

7.6 Connection to the four-witness Wick-rotation arc

Paper 42 [40] closes the four-witness Wick-rotation theorem (Hartle–Hawking [13], Sewell [28], Bisognano–Wichmann [3], Unruh [32]) at the operator-system level on the Riemannian truncated Camporesi–Higuchi spectral triple via the geometric BW-α and Tomita–Takesaki BW-γ modular Hamiltonians on a hemispheric wedge. Paper 43 [41] extends the closure to signature $(3, 1)$. Paper 45 [43] closes the $\mathcal{K}^+$-weak-form Lorentzian convergence; the present paper closes the strong-form Lorentzian convergence.

The triple $(42, 43, 45)$ together with Paper 46 constitute the operator-algebraic Lorentzian-extension closure of the math.OA quartet on the Riemannian side $(38, 39, 40)$. Both $\mathcal{K}^+$-weak-form and strong-form Lorentzian propinquity convergence are now closed, leaving only the enlarged-substrate question (G1$'$) as the deepest open question on this side.

7.7 Relation to Mondino–Sämann

Inherited from Paper 45 § 7.2: the synthetic Lorentzian Gromov–Hausdorff program of Mondino–Sämann [22] (and the downstream Che–Perales–Sormani [23] timed-Hausdorff extension) is conceptually adjacent ("Lorentzian Gromov–Hausdorff-type convergence") but operates on a categorically different mathematical object: pre-length spaces with causal diamonds, not operator-algebraic spectral triples. Neither construction implies the other; both are valid notions of Lorentzian-type Gromov–Hausdorff convergence on their respective categories.

7.8 Structural-skeleton scope

The strong-form convergence theorem obeys the structural-skeleton discipline of the GeoVac framework: the bound determines the rate, the asymptotic constant, and the limit identification on the full Krein-state-space substrate at finite cutoff. It does not determine calibration data (Yukawa selection, inner-factor parameters), which are external input. The selection of $T = 2\pi$ is structurally tied to the Bisognano–Wichmann modular period via Paper 43; the construction itself works for any $T > 0$.

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$n_{\max}$	Paper 45 $C_3^{(n_{\max})}$	Envelope $C_3^{\mathrm{op}}(n_{\max})$	N_{env}	$C_3^{(N_{\mathrm{env}})}$	Paper 45 underbinds?
2	0.5774	0.7071	3	0.7071	YES
3	0.7071	0.8165	5	0.8165	YES
4	0.7746	0.8660	7	0.8660	YES
5	0.8165	0.8944	9	0.8944	YES
10	0.9045	0.9487	19	0.9487	YES
20	0.9512	0.9747	39	0.9747	YES

Table 2: Envelope refinement at six $n_{\max}$ values. At every finite $n_{\max}$, Paper 45's stated $\sup_{N \le n_{\max}}$ under-states the bound required by the natural-substrate generators in the range $N \in (n_{\max}, 2n_{\max} - 1]$. The refined constant $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$ is the correct supremum value at the envelope.

A Envelope refinement of Paper 45 eq. (C3_joint_bound)

We record the finite-cutoff envelope refinement of Paper 45 eq. (C3_joint_bound) that arose from the L3b-2b sub-sprint analysis. The asymptotic statement of Paper 45 is unchanged; the refinement applies only to the explicit finite-$n_{\max}$ form of the per-harmonic Lichnerowicz constant on the natural Lorentzian substrate.

A.1 The Paper 45 statement and the refinement

Paper 45 eq. (C3_joint_bound) (in our notation):

$$C_3^{\mathrm{joint}}(n_{\max}, N_t) \;\le\; C_3^{\mathrm{SU}(2)}(n_{\max}) \;\le\; \sup_{2 \le N \le n_{\max}} \frac{N-1}{\sqrt{N^2-1}} \xrightarrow{n_{\max}\to\infty} 1^-.$$

The refined statement (this paper, Lemma 4.4):

$$C_3^{\mathrm{op}}(n_{\max}) \;\le\; \sup_{2 \le N \le 2n_{\max}-1} \frac{N-1}{\sqrt{N^2-1}} \;=\; \sqrt{1 - \frac{1}{n_{\max}}} \xrightarrow{n_{\max}\to\infty} 1^-.$$

Both forms tend to 1^- asymptotically; the refinement is a finite-cutoff correction.

A.2 Why Paper 45's range $N \le n_{\max}$ is non-binding on the natural substrate

The Avery–Wen–Avery $\Delta n = \pm 1$ ladder on $n_{\max}$ shells produces shell-difference labels N up to $2n_{\max} - 1$ (Paper 44 § 3). The natural-substrate multipliers $M^{\mathrm{spat}}_{N,L,M}$ in $\mathcal{O}^L_{n_{\max},N_t,T}$ include such labels in the range $N \in \{1, 2, \ldots, 2n_{\max} - 1\}$. At $n_{\max} = 3$, the per-harmonic constant at $N = 4$ is $C_3^{(4)} = 3/\sqrt{15} \approx 0.7746$, which exceeds Paper 45's stated $C_3^{(n_{\max}=3)} = 2/\sqrt{8} \approx 0.7071$. The natural-substrate generator ($N_{\mathrm{spat}} = 4, L = 1, M = \pm 1, p = 0$) has Lipschitz ratio $\approx 0.95 > C_3^{(3)} = 0.7071$ but $\le C_3^{(4)} = 0.7746$ (L3b-2b sub-sprint § 5).

The numerical witness:

A.3 Proposed Paper 45 erratum text

We propose the following minor refinement (NOT applied to Paper 45 in this paper; we record it here for completeness). Paper 45 eq. (C3_joint_bound) should be amended to read:

$$C_3^{\mathrm{joint}}(n_{\max}, N_t) \;\le\; C_3^{\mathrm{SU}(2)}(n_{\max}) \;\le\; \sup_{2 \le N \le 2n_{\max}-1} \frac{N-1}{\sqrt{N^2-1}} \;=\; \sqrt{1 - \frac{1}{n_{\max}}} \xrightarrow{n_{\max}\to\infty} 1^-.$$

The accompanying explanation: the supremum range is the Avery–Wen–Avery achievable-envelope cutoff $N \le 2n_{\max} - 1$ on the natural chirality-doubled scalar-multiplier substrate of the Lorentzian construction, NOT the smaller range $N \le n_{\max}$. The $\Delta n = \pm 1$ ladder on $n_{\max}$ shells produces shell-difference labels N up to $2n_{\max} - 1$ via Paper 44 § 3. Both forms tend to 1^- asymptotically; the asymptotic statement (Paper 45 Theorem 5.1) is unchanged. Numerical panel values in Paper 45 § 6 Table 1 are unchanged (they quote $\gamma^{\mathrm{SU}(2)}$ directly, not $C_3 \cdot \gamma^{\mathrm{SU}(2)}$).

A.4 Why Paper 38 does NOT need the analogous refinement

Paper 38 [37] § L3 Remark states $C_3(n_{\max}) \le \sup_{N \le n_{\max}} \sqrt{(N-1)/(N+1)} = \sqrt{(n_{\max}-1)/(n_{\max}+1)}$. Paper 38's Berezin map (Paper 38 eq. (B_def)) restricts to $N \le n_{\max}$ by Plancherel cutoff; the propinquity assembly (Paper 38 § 5.1 proof) bounds $\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \le C_3 \cdot \|\nabla f\|_\infty$ using the L3 bound on $N \le n_{\max}$ (the support of $B_{n_{\max}}(f)$). The natural substrate's full envelope $N \le 2n_{\max} - 1$ is NOT engaged in Paper 38's propinquity assembly; only Berezin-image multipliers with $N \le n_{\max}$ are.

Therefore Paper 38 § L3 Remark is correct as stated. The refinement is Lorentzian-substrate-specific: the chirality-doubling at the operator-system level admits more multipliers than the Berezin-image level, and the natural-substrate envelope extends to $N \le 2n_{\max} - 1$. Paper 38 has no chirality-doubled multiplier algebra (it lives in a single copy of the chirality-doubled $\mathcal{H}_{\mathrm{GV}}$), and the envelope and the Berezin-image support coincide.

B Strict-strong-form extension on the enlarged operator system

We record the closure of the named follow-on question (G1′, Sprint L3b-2f) from § 7.2: does the strong-form propinquity bound extend to chirality-flipping generators not present in the natural chirality-doubled scalar-multiplier substrate? This appendix shows that it does, with the substantive content that the chirality-flip enlargement is "paid" entirely in a gradient-norm extension — leaving the propinquity bound bit-exactly preserved at the same numerical value as the natural-substrate bound established in Theorem 5.1.

The arc that closes G1′ was internally executed as Sprint L3b-2f-β across four sub-sprints (a,b,c,d,e and a final β-L5 assembly). The structure of this appendix follows the β-L5 closure memo § 6.6.

B.1 Motivation

§ 7.2 named the enlarged-substrate question as the deepest open problem on the Lorentzian side of the GeoVac framework. The natural substrate $\mathcal{O}^L_{n_{\max},N_t,T}$ of Paper 44 restricts to multipliers commuting with the Krein fundamental symmetry J_L (Paper 44 Prop. 5.1), and equivalently to chirality-block-diagonal generators in the chirality-doubled basis of $\mathcal{H}_{\mathrm{GV}}$. The *strong-form* propinquity bound on this substrate is the content of Theorem 5.1. The natural-substrate "free upgrade" (§ 7) follows from the temporal-Lipschitz-invisibility identity (Lemma 3.2), which asserts that $[D_L^{\mathrm{diag}}, a] = 0$ identically for every $a \in \mathcal{O}^L$.

Two questions are then natural. (Q1′$_{\mathrm{a}}$) What happens on an enlarged substrate that adds chirality-flipping generators with $\{J_L, M\} = 0$? Lemma 3.2 fails on such generators: the cross term $[D_L^{\mathrm{diag}}, M \otimes M^{\mathrm{temp}}] \propto \{\gamma^0, M\} \otimes \partial_t M^{\mathrm{temp}}$ becomes non-zero, and the temporal direction becomes Lipschitz-relevant. (Q1′$_{\mathrm{b}}$) Does the asymptotic rate $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ survive on the enlarged substrate, possibly with a different multiplicative constant?

The closure below answers both questions affirmatively, with the chirality-flip content absorbed into the gradient-norm side of the L3 bound and the propinquity multiplier $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$ unchanged.

B.2 The enlarged operator system and the J-graded gradient norm

Definition B.1 (Enlarged operator system)**.** Define the *enlarged* natural-substrate operator system

$$\mathcal{O}^L_{\text{enlarged}} \;:=\; \mathcal{O}^L_{n_{\max},N_t,T} \cup \{M^{\text{spat,flip}}_{N,L,M} \otimes M^{\text{temp}}_p : \; N \le n_{\max},\; |p| \le K_{\max}\}^*, \tag{27}$$

generated as a $*$-closed subspace of $B(\mathcal{K}_{n_{\max},N_t})$, where the chirality-flip spatial multiplier is the chirality-asymmetric doubling

$$M^{\text{spat,flip}}_{N,L,M} \;:=\; \operatorname{diag}\big(W^{NLM}_W,\; -W^{NLM}_W\big) \tag{28}$$

in the chiral basis of $\mathcal{H}_{\text{GV}}$, with W^{NLM}_W the Avery–Wen–Avery Weyl-spinor multiplier of Paper 44 § 3.

The relation between this and the off-block-diagonal form of eq. (26) deserves attention.

Remark B.2 (Convention distinction: minimal anti-commuting enlargement)**.** § 7.2 wrote a candidate enlargement using the off-block-diagonal form $M_{\text{flip}} = \left(\begin{smallmatrix} 0 & W \\ W^* & 0 \end{smallmatrix}\right)_\chi$. The convention in Paper 44 sets $\gamma^0 = \left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$ (chiral basis). With this convention, for Hermitian W,

$$\{\gamma^0, \left(\begin{smallmatrix} 0 & W \\ W^* & 0 \end{smallmatrix}\right)\} \;=\; \left(\begin{smallmatrix} W^*+W & 0 \\ 0 & W+W^* \end{smallmatrix}\right) \;\neq\; 0 \text{ in general,}$$

so the off-block-diagonal form does not in general anti-commute with γ^0. In contrast, the chirality-asymmetric diagonal form (28) satisfies

$$\{\gamma^0, \operatorname{diag}(W, -W)\} \;=\; \left(\begin{smallmatrix} 0 & -W+W \\ W-W & 0 \end{smallmatrix}\right) \;=\; 0 \qquad \text{bit-exact.}$$

The chirality-asymmetric diagonal form is therefore the correct *minimal anti-commuting* enlargement in the Paper 44 chiral basis; § 7.2's heuristic description used a convention-loose form and is sharpened here by (28). Both forms yield non-zero $\{\gamma^0, M\}$ on generic choices, but only (28) gives the bit-exact anti-commutation that the proofs below rely on.

The natural-substrate Lipschitz seminorm L_{op} no longer captures the full Lipschitz content of multipliers in $\mathcal{O}^L_{\text{enlarged}}$: the temporal commutator $[D^{\text{diag}}_L, M^{\text{spat,flip}} \otimes M^{\text{temp}}]$ is now non-zero, and a strong-form L3 bound must account for it. The L3 derivation of § 4.3 relies on the temporal-Lipschitz-invisibility identity; on the enlarged substrate this identity must be replaced by a *splitting*.

Definition B.3 (J-graded gradient norm)**.** Let $f \in C^\infty(S^3 \times S^1_T)$, and decompose its joint Peter–Weyl $\times$ Fourier expansion into J_L-commuting and J_L-anti-commuting components,

$$f \;=\; f^{\text{block}} + f^{\text{flip}}, \qquad J f^{\text{block}} J^{-1} = f^{\text{block}}, \quad J f^{\text{flip}} J^{-1} = -f^{\text{flip}}, \tag{29}$$

where the splitting acts at the level of the spatial multiplier component (the temporal factor M^{temp}_q commutes with J_L automatically). Define the *J_L-graded gradient norm*

$$G^{\text{enlarged}}(f) \;:=\; \|\nabla^{\text{joint}} f^{\text{block}}\|_\infty \;+\; 2\,\|D^{\text{diag}}_L\|_{\text{op}} \cdot \|f^{\text{flip}}\|_\infty. \tag{30}$$

The structural interpretation of the second term in eq. (30) is that the temporal direction $D^{\text{diag}}_L = i\gamma^0 \otimes \partial_t$ becomes Lipschitz-relevant on chirality-flipping generators. The prefactor 2 reflects the anti-commutation $\{\gamma^0, M^{\text{spat,flip}}\} = 0$, which forces the cross term in the commutator $[D^{\text{diag}}_L, M^{\text{spat,flip}} \otimes M^{\text{temp}}]$ to combine additively rather than cancel (L3b-2a sub-sprint § 3). The operator norm $\|D^{\text{diag}}_L\|_{\text{op}} = \|\partial_t\|_{\text{op},(N_t,T)} = O(N_t/T)$ on the truncated S^1_T is finite and uniformly bounded over the verification panel; we treat it as a known constant at each panel cell.

Remark B.4 (Restriction to natural substrate)**.** On the natural substrate $\mathcal{O}^L \subset \mathcal{O}^L_{\text{enlarged}}$, every multiplier is J_L-commuting, so $f^{\text{flip}} = 0$ and $G^{\text{enlarged}}(f) = \|\nabla^{\text{joint}} f^{\text{block}}\|_\infty = \|\nabla^{\text{joint}} f\|_\infty$. The enlarged gradient norm reduces to the natural gradient norm bit-exactly, and the L3 bound on the enlarged substrate reduces to the L3 bound of Lemma 4.4 bit-exactly. The enlarged construction is a strict extension of the natural construction; no natural-substrate result is altered.

B.3 Five lemmas transport under the enlarged substrate

We transport the five-lemma structure of § 4 to the enlarged substrate under the enlarged gradient norm G^{enlarged}.

L1′ (substrate). The propagation number of the enlarged substrate is

$$\text{prop}_{\text{achievable}}(\mathcal{O}^L_{\text{enlarged}}) \;=\; 1. \tag{31}$$

The enlarged substrate is a denser generating set than the natural substrate's prop = 2: chirality-asymmetric content provides cross-block coupling between $\mathcal{K}^+$ and $\mathcal{K}^-$ that closes the achievable envelope in one application of the multiplier algebra (L3b-2a sub-sprint § 4). Concretely, every element of $\mathcal{O}^L_{\text{enlarged}}$ can be written as a sum of products of two generators from $\mathcal{O}^L_{\text{enlarged}}$, including off-diagonal Krein-block matrix elements which the natural substrate cannot reach in one step.

The lemma statement is otherwise unchanged from Paper 44: $\mathcal{O}^L_{\text{enlarged}}$ is unital $*$-closed, D_L is Krein-self-adjoint on the enlarged substrate (the same D_L acts on the enlarged Krein space), and J_L is a fundamental symmetry of square $+I$. The condition $[J_L, a] = 0$ no longer holds for every $a \in \mathcal{O}^L_{\text{enlarged}}$; instead, each a splits as in eq. (29) with a J_L-block part and a J_L-flip part. At $N_t = 1$, the enlarged substrate reduces bit-exactly to the Riemannian chirality-doubled enlarged truncated spectral triple on $\mathcal{H}^{n_{\max}}_{\text{GV}}$.

L2 (joint cb-norm). The joint cb-norm is seminorm-independent: it depends only on the Berezin–Plancherel structure of the central spectral Fejér kernel $K^{\text{joint}} = K^{\text{SU}(2)} \otimes K^{U(1)}$, not on the choice of Lipschitz seminorm on operators. Lemma 4.2 therefore transports verbatim to the enlarged substrate:

$$\|S_{K^{\text{joint}}_{n_{\max},N_t,T}}\|_{\text{cb}} \;=\; \frac{2}{n_{\max}+1}, \qquad N_t\text{-independent.} \tag{32}$$

The cb-norm controls the propinquity height contribution from the Berezin map (Paper 45 Remark 4.4), independently of which substrate the Berezin image targets. See Paper 46 § 4.2 and Paper 45 § 4.2.

L3 (joint Lichnerowicz). On the enlarged substrate, the joint Lichnerowicz comparison takes the form

$$\|[D_L, B^{\text{joint}}_{\text{enlarged}}(f)]\|_{\text{op}} \;\le\; C_3^{\text{op}}(n_{\max}) \cdot G^{\text{enlarged}}(f), \tag{33}$$

with $C_3^{\text{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$ as in eq. (14). The numerical constant is unchanged from the natural-substrate L3 bound (Lemma 4.4). The cost of the chirality-flip enlargement is absorbed entirely into the gradient-norm extension $G^{\text{enlarged}}(f)$ via eq. (30).

Sketch. Split $f = f^{\text{block}} + f^{\text{flip}}$ and correspondingly $B^{\text{joint}}_{\text{enlarged}}(f) = B^{\text{joint}}(f^{\text{block}}) + B^{\text{joint}}_{\text{flip}}(f^{\text{flip}})$, with $B^{\text{joint}}_{\text{flip}}$ the Berezin map valued in the chirality-flip sector. For the block part, Lemma 3.2 applies verbatim and gives $\|[D_L, B^{\text{joint}}(f^{\text{block}})]\|_{\text{op}} \le C_3^{\text{op}}(n_{\max}) \cdot \|\nabla^{\text{joint}} f^{\text{block}}\|_\infty$ by Lemma 4.4. For the flip part, $[D_L, B^{\text{joint}}_{\text{flip}}(f^{\text{flip}})] = [D_L^{\text{diag}}, \cdot] + [D_L^{\text{off}}, \cdot]$, with both pieces non-zero on flip generators. The $[D_L^{\text{off}}, \cdot]$ piece is bounded by the spatial commutator norm in the usual way (Lemma 4.4 sketch, step C, transported to flip generators). The $[D_L^{\text{diag}}, \cdot]$ piece is bounded by $\|[D_L^{\text{diag}}, M^{\text{spat,flip}} \otimes M^{\text{temp}}]\|_{\text{op}} \le 2\,\|D_L^{\text{diag}}\|_{\text{op}} \cdot \|M^{\text{spat,flip}}\|_{\text{op}} \cdot \|M^{\text{temp}}\|_{\text{op}}$, where the factor 2 arises from $\{\gamma^0, M^{\text{spat,flip}}\} = 0$ (anti-commutation combines additively). The triangle inequality and the structural sup-bound $\|M^{\text{spat,flip}}_{N,L,M}\|_{\text{op}} \le 1$ on the Avery–Wen–Avery normalized basis give the total flip-piece bound $2\,\|D_L^{\mathrm{diag}}\|_{\mathrm{op}} \cdot \|f^{\mathrm{flip}}\|_\infty$, which is the second term of $G^{\mathrm{enlarged}}(f)$ in eq. (30). Combining,

$$\begin{aligned}
\|[D_L, B^{\mathrm{joint}}_{\mathrm{enlarged}}(f)]\|_{\mathrm{op}} &\le \|[D_L, B^{\mathrm{joint}}(f^{\mathrm{block}})]\|_{\mathrm{op}} + \|[D_L, B^{\mathrm{joint}}_{\mathrm{flip}}(f^{\mathrm{flip}})]\|_{\mathrm{op}} \\
&\le C_3^{\mathrm{op}}(n_{\max}) \cdot \|\nabla^{\mathrm{joint}} f^{\mathrm{block}}\|_\infty + C_3^{\mathrm{op}}(n_{\max}) \cdot 2\|D_L^{\mathrm{diag}}\|_{\mathrm{op}} \cdot \|f^{\mathrm{flip}}\|_\infty \\
&= C_3^{\mathrm{op}}(n_{\max}) \cdot G^{\mathrm{enlarged}}(f).
\end{aligned}$$

The penultimate line uses the structural fact that the $[D_L^{\mathrm{off}}, \cdot]$ contribution from the flip part is dominated by the same Avery–Wen–Avery envelope as the block part (the flip multiplier supports overlap the block supports), so the spatial-commutator bound applies with the same $C_3^{\mathrm{op}}(n_{\max})$. See the β-L5 closure memo § 6.4 for the full step-by-step derivation.

L4 (joint Berezin reconstruction). The joint Berezin map extends to the enlarged substrate as $B^{\mathrm{joint}}_{\mathrm{enlarged}} : C^\infty(S^3 \times S^1_T) \to \mathcal{O}^L_{\mathrm{enlarged}}$ via the J_L-splitting of f at the spatial multiplier level. Properties (a)–(c), (e) of Lemma 4.8 transport verbatim by the seminorm-independence remark (Remark 4.9): positivity, contractivity, approximate identity, and Krein-grading preservation do not engage the choice of Lipschitz seminorm. Property (d) (L3 compatibility) transports under the enlarged gradient norm: for every $f \in C^\infty(S^3 \times S^1_T)$,

$$\|[D_L, B^{\mathrm{joint}}_{\mathrm{enlarged}}(f)]\|_{\mathrm{op}} \le C_3^{\mathrm{op}}(n_{\max}) \cdot G^{\mathrm{enlarged}}(f), \tag{34}$$

by direct combination of Lemma 4.8(d) (block part) and eq. (33) (full bound under G^{enlarged}). The approximate-identity rate $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ is unchanged: Paper 45 § 4.3(c) and Lemma 4.8(c) factor the convolution deficit through smooth-gradient norms of f, and the J_L-splitting of f does not affect the convolution-deficit bound. Each split component is independently smooth and contributes its own smooth-gradient piece, recombining into $\nabla^{\mathrm{joint}} f$.

L5 (Latrémolière propinquity assembly). The propinquity assembly transfers under the enlarged gradient norm with the same multiplier constant $C_5^{\mathrm{joint}} = 1$ as in § 4.5. The reach and height contributions of the joint tunneling pair $(B^{\mathrm{joint}}_{\mathrm{enlarged}}, P^{\mathrm{joint}})$ are bounded by the enlarged-substrate analogs of eqs. (18) and (34). Concretely,

$$\mathrm{reach}_B^{\mathrm{enlarged}} \le \gamma^{\mathrm{joint}}_{n_{\max},N_t,T}, \qquad \mathrm{height}_B^{\mathrm{enlarged}} \le C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T},$$

and the assembled propinquity bound is

$$\Lambda^{\mathrm{strong}}\big(\mathcal{T}^{L,\mathrm{enlarged}}_{n_{\max},N_t,T},\ \mathcal{T}^L_{S^3\times S^1_T}\big) \le C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T}, \tag{35}$$

with C_3^{op} and γ^{joint} identical to those of Theorem 5.1. The temporal-Lipschitz content of the chirality-flip enlargement is absorbed into the gradient-norm side $G^{\mathrm{enlarged}}(f)$ of the L3 / L4 bounds, leaving the propinquity multiplier unchanged.

B.4 Main statement: strict-strong-form extension

Theorem B.5 (Strict-strong-form extension, β-L5 closure). *On the enlarged operator system $\mathcal{O}^L_{\mathrm{enlarged}} = \mathcal{O}^L \cup \{M^{\mathrm{spat,flip}} \otimes M^{\mathrm{temp}}\}^*$ of Definition B.1, with the J_L-graded gradient norm G^{enlarged} of Definition B.3 replacing the natural-substrate joint smooth gradient norm, the Latrémolière propinquity bound is*

$$\Lambda^{\mathrm{strong}}\big(\mathcal{T}^{L,\mathrm{enlarged}}_{n_{\max},N_t,T},\ \mathcal{T}^L_{S^3\times S^1_T}\big) \le C_3^{\mathrm{op}}(n_{\max}) \cdot \gamma^{\mathrm{joint}}_{n_{\max},N_t,T} \xrightarrow[(n_{\max},N_t)\to(\infty,\infty)]{} 0, \tag{36}$$

with $C_3^{\mathrm{op}}(n_{\max}) = \sqrt{1 - 1/n_{\max}}$ and $\gamma^{\mathrm{joint}} = O(\log n_{\max}/n_{\max} + T/N_t)$ identical to those of Theorem 5.1. The Riemannian-limit recovery at $N_t = 1$ is bit-exact (Frobenius residual 0.0 in float64).

Proof. Combine the five-lemma transport of § B.3. L1′ supplies the enlarged operator-system substrate with $\mathrm{prop}_{\mathrm{achievable}} = 1$. L2 supplies the joint cb-norm seminorm-independent bound $2/(n_{\max}+1)$. L3 supplies eq. (33) on the Berezin image. L4 supplies the approximate-identity rate γ^{joint} and the L3 compatibility under the enlarged gradient norm (eq. (34)). L5 assembles the bound (eq. (35)) via the joint tunneling pair. The Riemannian-limit recovery follows from the natural-substrate limit (Theorem 5.1 F1) together with the observation that the chirality-flip enlargement reduces bit-exactly to a chirality-flip enlargement of the Riemannian chirality-doubled truncation at $N_t = 1$, on which the same gradient-norm splitting applies and the same $C_3^{\mathrm{op}}(n_{\max})$ bound holds. □

B.5 Interpretation: the enlargement is paid in the gradient norm

The substantive content of Theorem B.5 is that the enlarged-substrate bound is bit-exactly equal to the natural-substrate bound of Theorem 5.1, at the same C_3^{op} and the same γ^{joint}, on the same numerical panel $(n_{\max}, N_t) \in \{(2,3),(3,5),(4,7)\}$ with values 2.0746, 1.6101, 1.3223. The cost of the chirality-flip enlargement is absorbed entirely into the gradient-norm side via eq. (30):

$$G^{\mathrm{enlarged}}(f) - \|\nabla^{\mathrm{joint}} f^{\mathrm{block}}\|_\infty \;=\; 2\|D_L^{\mathrm{diag}}\|_{\mathrm{op}} \cdot \|f^{\mathrm{flip}}\|_\infty.$$

This is the only place where chirality-flip content appears in the extended construction. Three structural readings of this fact are useful.

Reading 1: the enlargement is "paid" in f-space, not operator space. The operator-side multiplier constant C_3^{op} and the convergence-rate moment γ^{joint} are unchanged. The extension cost is borne by the function-space norm of f, which grows by an additive $2\|D_L^{\mathrm{diag}}\|_{\mathrm{op}} \cdot \|f^{\mathrm{flip}}\|_\infty$ term reflecting the temporal Lipschitz content of the chirality-flip sector.

Reading 2: the natural-substrate "free upgrade" is the $f^{\mathrm{flip}} = 0$ specialization. On the natural substrate ($\mathcal{O}^L \subset \mathcal{O}^L_{\mathrm{enlarged}}$), every multiplier is J_L-commuting, so the chirality-flip content of f is zero, and $G^{\mathrm{enlarged}}(f) = \|\nabla^{\mathrm{joint}} f\|_\infty$ bit-exact. The enlarged construction reduces to the natural construction precisely when $f^{\mathrm{flip}} = 0$, and Theorem 5.1 is the $f^{\mathrm{flip}} = 0$ specialization of Theorem B.5.

Reading 3: the L3b-2a temporal-Lipschitz-invisibility identity "breaks" on enlarged at exactly the rate predicted by $\|D_L^{\mathrm{diag}}\|_{\mathrm{op}}$. The breakdown is quantitatively controlled: the new term $2\|D_L^{\mathrm{diag}}\|_{\mathrm{op}} \cdot \|f^{\mathrm{flip}}\|_\infty$ is bounded uniformly over each panel cell, and over the full $n_{\max} \to \infty$ limit the chirality-flip contribution does not qualitatively alter the asymptotic-rate analysis of § 4.5.

The closure also clarifies the structural status of the G1′ question in § 7.2. The original internal Sprint L3b-2f prediction was that the enlarged-substrate bound would exceed the natural bound "by a finite factor of order $\sqrt{2}$ to 2, the chirality-flipping content roughly doubles the operator-norm content". The β-L5 closure refines this prediction to "exceeds by exactly zero in the operator-side bound $C_3^{\mathrm{op}} \cdot \gamma^{\mathrm{joint}}$; the chirality-flipping content is absorbed into the gradient-norm extension G^{enlarged}". The structural reason is that the chirality-flip enlargement preserves the Avery–Wen–Avery achievable envelope on $n_{\max}$ shells (the chirality-asymmetric doubling reuses the same $N \le 2n_{\max} - 1$ range as the block-diagonal multipliers), and the Lichnerowicz bound of Lemma 4.4 applies with the same envelope constant on both block and flip sectors. The L3b-2a / 2b / 2c / 2d sub-sprints established this envelope-preservation explicitly; the β-L5 sub-sprint assembled the result.

B.6 Status of G1′ and the strict-strong-form closure

Theorem B.5 closes G1′ (§ 7.2) in the strict-strong-form sense: the strong-form Lorentzian propinquity bound extends from the natural substrate to the enlarged operator system $\mathcal{O}^L_{\mathrm{enlarged}}$ at the same numerical constant and the same asymptotic rate, with the chirality-flip content absorbed into the J_L-graded gradient norm G^{enlarged}. The G1′ question is therefore answered: the enlarged-substrate bound equals the natural-substrate bound at the operator-side constant C_3^{op} and the convergence-rate moment γ^{joint}.

The remaining open questions on the Lorentzian side of the GeoVac framework are unchanged by this appendix: G2 (de-compactification $T \to \infty$, § 7.3), G3 (cross-manifold extensions, § 7.4), and G4 (inner-factor calibration data, § 7.5). With G1′ now closed, the operator-algebraic Lorentzian-extension closure of the math.OA quartet (Papers 42, 43, 45, 46) is structurally complete on the natural and enlarged substrates at finite cutoff; the open questions are confined to limit-taking ($T \to \infty$), cross-manifold extensions, and inner-factor calibration, all of which are orthogonal to the present strong-form convergence theorem.

The closure result here strengthens the connection to the four-witness Wick-rotation arc of § 7.6: Papers 42, 43 close the modular structure on the natural-substrate hemispheric wedge, Paper 45 closes the $\mathcal{K}^+$-weak-form convergence, the body of Paper 46 closes the strong-form on the natural substrate, and Theorem B.5 extends the strong-form to the enlarged operator system. The chirality-flip content of operator algebras on Lorentzian Krein spectral triples at finite cutoff is now within the framework of Latrémolière propinquity convergence.

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